Response to Reviewer Comments

Context Without Contact:

Top-Down Information Flow from Shared Modulation in Uncoupled Prebiotic Systems

Manuscript submitted to Royal Society Open Science

Celia Blanco

celia.blanco@bmsis.com

We appreciate the reviewers' thoughtful and constructive feedback, which has helped strengthen the manuscript. We have carefully considered all comments and addressed each point in the revised version. Below we provide a detailed, point-by-point response describing the changes made. Revisions in the manuscript and supplementary information are indicated in red.

Reviewer: 1

Comments to the Author(s)

I think the ideas put forward by the Author are valuable and worth exploring, but I recommend

major revisions (see attached PDF).

Major issues

1. The article is structured into three sections: the Introduction provides background, the Results section outlines the methodology and numerical experiments, and the Discussion interprets the findings in the context of the origin of life. I recommend a more thorough treatment of the underlying themes. In particular, the Introduction would benefit from additional detail and references to prior work that has considered top-down causation in the context of the origin of life and evolutionary transitions. Greater attention should also be given to the role of logistic map models in computational biology and the application of information-theoretic measures. These additions would substantially strengthen the Introduction, which in its current form is insufficient.

We have added three new paragraphs to the Introduction:

(a) Top-down causation in origin of life and evolutionary transitions

We added a paragraph reviewing prior work on top-down causation in major evolutionary transitions and the origin of life. It summarizes the taxonomy of top-down causation mechanisms (Ellis 2012), the integration of previously independent replicators into higher-level units (Szathmáry & Maynard Smith

1995; Szathmáry 2015), and top-down causation by information control (Auletta, Ellis & Jaeger 2008). It also notes the characterization of the origin of life as a transition from dynamics governed by physical law to those governed by information (Walker 2014; Walker & Davies 2013), and introduces directional information flow between scales as an operational framework (Walker 2012).

> A taxonomy of top-down causation mechanisms has been proposed, distinguishing algorithmic, non-adaptive, and adaptive forms, with the case made that downward causation is essential for understanding biological organization.
>
> […]
>
> One way to study such transitions is by examining directional information flow between scales, specifically whether the macroscopic state of a system can predict the dynamics of its parts better than the reverse.

(b) Logistic maps in computational biology

We added a paragraph that provides context for the use of logistic map models in population dynamics and nonlinear systems. The paragraph notes their role as minimal models exhibiting deterministic chaos (May 1976; Strogatz 2024), acknowledges that synchronization of chaotic systems under common external forcing is a well-established phenomenon in nonlinear dynamics (Boccaletti et al. 2002), and cites prior work using coupled logistic maps to study collective behavior and information flow (Cisneros et al. 2002; Kaneko 1990). We also note that, in origins-of-life research, abstract dynamical models are commonly used to isolate general principles from biochemical complexity (Ruiz-Mirazo et al. 2014), motivating the use of a minimal model in the present study.

> To investigate this question, we employ logistic maps as our model system. The logistic map, originally introduced as a minimal model of population dynamics, has become a paradigmatic tool in nonlinear dynamics and computational biology.
>
> […]
>
> In the context of prebiotic chemistry, such minimal (albeit abstract) models can be extremely valuable because they isolate the dynamical consequences of environmental forcing from the biochemical complexity of real systems, allowing the identification of generic mechanisms that do not depend on molecular detail

(c) Information-theoretic measures

We added a paragraph reviewing the application of transfer entropy (Schreiber 2000) across fields, including neuroscience (Vicente et al. 2011; Bossomaier et al. 2016), climate science (Runge et al. 2014; Runge 2018), ecology (Sugihara et al. 2012), and multi-scale biological systems (Walker 2012; Cisneros et al. 2002). The paragraph also discusses TE's sensitivity to common-driver confounds (Lizier & Prokopenko 2010; Runge et al. 2014), and introduces conditional transfer entropy (CTE) as a principled correction (Smirnov & Bezruchko 2012; Runge 2018), which we employ to separate effects of shared environmental forcing from residual population-level structure.

TE has become a standard tool for detecting directed information flow in complex systems and has been applied extensively in neuroscience to map effective connectivity between brain regions.

[…]

We employ both unconditional and conditional TE in our analysis to distinguish directional predictability arising from shared environmental forcing from residual population-level structure.

2. I strongly recommend adding a dedicated methodology section that clearly presents the model and parameter choices, some of which appear arbitrary (e.g., r_min, r_max, K_0, A, ω). This would help demonstrate the robustness of the argument. The Author may also consider reorganizing the article into ad hoc sections corresponding to the different stages of the analysis, though this is left to their discretion.

We have reorganized the manuscript and added a dedicated Methods section to clarify the model and parameter choices. This section now (i) provides the explicit equations used in the analysis for the information-theoretic measures, including transfer entropy (TE) and conditional transfer entropy (CTE); (ii) presents the formulas for the regression-based alignment metrics (R^2 and SNR); (iii) explains the model parameters and their rationale (including r_{min}, r_{max}, K_0, A, and ω); and (iv) describes the analysis procedure, including TE estimations and the surrogate significance test.

We have also reorganized the Results section into a set of subsections corresponding to the different stages of the analysis.

3. From a methodological standpoint, I am not convinced that the Author demonstrates convincingly that the alignment of individual units with the mean field is due to top-down causation rather than to forcing caused by the environmental modulation A. [...] I encourage the Author to supplement their analysis by isolating constraint-driven order from mean-field-driven order. Conditional transfer entropy could be useful for this purpose. Moreover, I encourage the Author to pay particular attention to the value of A/K_0, which determines the extent to which the quadratic term influences the forcing from the environmental constraint. Combined with a deeper analysis of Figure 1 — for instance, why TE increases monotonically with A, and why the confidence interval rises sharply when A ≈ 50 — this would substantially strengthen the Author's argument.

We thank the reviewer for this important suggestion, which has substantially strengthened the manuscript. We have addressed all three sub-points:

(a) Conditional transfer entropy (CTE).

We added a new Results subsection titled *Conditional transfer entropy reveals macro–micro asymmetry*. We implemented a conditional transfer entropy analysis in which TE was computed in both directions ($M \rightarrow x$ and $x \rightarrow M$) while conditioning on the environmental signal K_n across the same amplitude sweep as

in Figure 1B. The results are presented in a new figure (Figure 3). Conditioning substantially reduces the magnitude of TE, however, a residual asymmetry remains, and CTE($M \rightarrow x \mid K_n$) exceeds CTE($x \rightarrow M \mid K_n$) at moderate to high amplitudes. We therefore interpret the TE asymmetry as reflecting two components: environmentally mediated coordination and a smaller residual structure arising from nonlinear aggregation of the population dynamics.

> **Conditional transfer entropy reveals macro–micro asymmetry**
>
> The unconditional TE asymmetry demonstrated above could, in principle, arise entirely from the shared environmental signal K_n acting as a common driver.
>
> […]
>
> The results from the conditional analysis indicate that the macro-micro information structure observed in the unconditional analysis has two components: (1) a dominant contribution arising from the shared environmental modulation, and (2) a directional asymmetry that emerges from the collective dynamics of the population.

We also added a paragraph to the Discussion section:

> The conditional transfer entropy (CTE) analysis clarifies the source of this asymmetry.
>
> [...]
>
> Our results therefore suggest that the TE asymmetry reflects two complementary mechanisms: environmentally scaffolded coordination, which dominates under strong forcing, and a weaker emergent component arising from nonlinear aggregation of the population dynamics.

And we added a sentence to the abstract:

> Conditional transfer entropy (CTE), conditioning on the environmental signal, reveals that a macro-to-micro asymmetry remains even after the influence of the shared driver is accounted for. This persistence suggests that nonlinear aggregation of uncoupled chaotic units generates predictive structure beyond the common environmental forcing.

(b) Role of the ratio A/K_0.

Following the reviewer's suggestion, we performed additional simulations across multiple baseline carrying capacities (K_0 = 50, 100, 200, 400). The resulting curves collapse when plotted as a function of A/K_0, confirming that this ratio acts as the relevant control parameter for the strength of environmental forcing. These results are presented in a new Supplementary Figure 2 and discussed in the Results section.

> Because the modulation enters the dynamics through the carrying capacity $K_n = K_0 + A\sin(\omega n)$, the effective strength of the forcing is determined by the ratio A/K_0. Repeating the simulations for different baseline carrying capacities (K_0 = 50, 100, 200, 400) shows that the TE($M \rightarrow x$) curves

collapse when plotted as a function of A/K_0 (Supplementary Figure S2), indicating that the relative modulation amplitude acts as the relevant control parameter governing the growth of transfer entropy. The reverse, TE($x \to M$), remains near zero across A/K_0 values.

(c) Interpretation of Figure 1B.

We added explicit explanations in the Results section addressing the two features noted by the reviewer. First, the monotonic increase of TE($M \to x$) with A arises because larger modulation amplitudes strengthen the shared environmental component relative to unit-specific chaotic fluctuations, making individual trajectories increasingly aligned with the population mean.

The monotonic increase of TE($M \to x$) arises because larger modulation amplitudes strengthen the shared environmental component relative to unit-specific chaotic fluctuations, causing individual trajectories to become progressively more aligned with the mean field. In contrast, the reverse TE ($x \to M$) remains small and tends to decrease with increasing modulation amplitude, reflecting the fact that the population mean increasingly averages out unit-specific fluctuations.

Second, the widening confidence intervals at large amplitudes (A ~ 40-50) reflect increased sensitivity to initial conditions as the forcing approaches the scale of the baseline carrying capacity (A/K_0 ~ 0.5), which amplifies seed-to-seed variability in the chaotic trajectories.

The widening of the confidence interval (Figure 1B) at large amplitudes ($A \gtrsim 40 - 50$) reflects increased variability across simulation as the modulation amplitude approaches the scale of the baseline carrying capacity ($A/K_0 \sim 0.5$). In this regime, the strong environmental forcing interacts nonlinearly with the heterogeneous chaotic dynamics of the units, making the resulting trajectories more sensitive to differences in initial conditions and parameter realizations.

Minor issues

Please unify the terminology: "ensemble mean" (p. 4) vs. "mean field" (p. 5), "population mean" (p. 6), and so on.

We have standardized the terminology to mean field throughout the entire manuscript (main text, figure captions, and supplementary materials). All instances of “ensemble mean” and “population mean” have been replaced.

p. 6: In Figure 1B, I suggest adding an inset illustrating how TE varies as a function of ω.

We have added a new Supplementary Figure S1 showing TE(M→x) as a function of the modulation period (2π/ω) at fixed amplitude A = 25. We also added a brief description to the Results section:

> Transfer entropy from the mean field to individual units, TE$(M \to x)$, remains high for slow modulation (long periods) but decreases as the modulation frequency increases (Supplementary Figure S1). This behavior indicates that very rapid environmental oscillations average out relative to the intrinsic chaotic dynamics of the units, preventing the forcing from imposing a coherent population-wide structure. The reverse direction TE$(x \to M)$ remains near zero across modulation periods.

and to the Methods section:

> The modulation frequency $\omega = 2\pi/2000$ was chosen to provide approximately 4.5 full modulation cycles within the analysis window (robustness to ω is demonstrated in Supplementary Figure S1).

p. 7: The Author writes that "The saturation was not due to numerical clipping; although a soft bound..." Please explain.

We have expanded this passage to clarify three points: (a) what the soft bound is; (b) how often the bound was approached; and (c) why this rules out clipping as the driver of TE saturation:

> The saturation was not due to numerical clipping; although a soft bound of (0, $2K_0$) = (0, 200) was imposed on all unit states at each time step to prevent unphysical negative concentrations and runaway growth. At the highest modulation amplitude tested (A = 60), the lower bound ($x \leq 0$) was approached in fewer than 10% of unit-timestep pairs across all seeds, and the upper bound ($x \geq 200$) was never reached. Clipping, therefore, cannot account for the TE saturation.

p. 7: Please explicitly write out the SNR and R^2 formulas actually computed.

We have added formal variable definitions and explicit formulas for both metrics in the new Methods section under *Regression-based alignment metrics*.

p.11: The address of the GitHub repo is incorrect.

We thank the reviewer for catching this. The GitHub repository URL has been corrected in both the main manuscript and the supplementary information (dashes were mistakenly used instead of underscores).

Supplementary Information, p. 4: The caption should indicate "Return maps of [...] A = {0, 20, 40, 60}".

We thank the reviewer (yet again!) for catching this. The caption for Supplementary Figure S5 has been corrected to indicate A = {0, 20, 40, 60}, matching the actual figure panels.

Reviewer: 2

Comments to the Author(s)

The starting point of this work is the fact that a set of chaotic systems can synchronize even in the absence of mutual coupling when driven by a common external signal, for example, through the modulation of one of their control parameters. This is a well-known phenomenon, characterized in the classical literature on chaotic systems. [...] I think this work should be mentioned, in particular for its section 5: Boccaletti, S., Kurths, J., Osipov, G., Valladares, D. L., & Zhou, C. S. (2002). The synchronization of chaotic systems. Physics Reports, 366(1–2), 1–101.

We thank the reviewer for this important reference. We now explicitly acknowledge in the Introduction that the synchronization of chaotic systems under common external forcing is a well-characterized phenomenon in nonlinear dynamics, citing Boccaletti et al. (2002).

> More broadly, the synchronization of chaotic systems under common external forcing is a well-characterized phenomenon in nonlinear dynamics (Boccaletti et al. 2002); even without mutual coupling,
>
> […]
>
> such minimal models are valuable because they isolate the dynamical consequences of environmental forcing from the biochemical complexity of real systems, allowing the identification of generic mechanisms that do not depend on molecular detail (Ruiz-Mirazo et al. 2014).

It goes without saying that it would be appreciated, at least in the continuation of this research line, to show the implications of the concept of synchronization in a simulation including some concrete aspect of biochemistry or biophysics.

We agree with the reviewer that exploring these ideas in models incorporating explicit biochemical or biophysical processes would be a valuable next step. While this lies beyond the scope of the present study, we have added a brief statement at the end of the Discussion noting this as a natural direction for future work. We hope this addition helps motivate further work along these lines.

> Future work could test these principles in models incorporating more realistic prebiotic chemistry or biophysics—for example, protocell populations with explicit metabolic kinetics evolving under cyclic environmental forcing—and evaluate whether macro-micro information asymmetries arise when information flow is measured between collective chemical states and individual protocell dynamics.

Reviewer: 3

Comments to the Author(s)

This is a clear and well-executed manuscript that addresses an important conceptual question: whether directional informational asymmetry can arise purely from shared environmental structure in the absence of internal coupling. The modeling choices are appropriately minimal, and the interpretation of transfer entropy as an operational measure of predictability rather than mechanistic causation is careful and convincing.

One minor suggestion would be to sharpen, perhaps in the Discussion, the distinction between externally imposed boundary conditions and internally generated coordination, to further preempt potential misunderstandings regarding “top-down” causation. Otherwise, I believe the manuscript is suitable for publication with only minor clarifications.

We thank the reviewer for this helpful suggestion. We have added a paragraph in the Discussion explicitly distinguishing externally imposed boundary conditions from internally generated coordination. The new text clarifies that the “top-down” signal observed in our model arises from shared environmental constraints rather than from internally coupled dynamics, helping to avoid potential misunderstandings about the interpretation of top-down causation.

> It is important to distinguish two modes of coordination that are often conflated under the umbrella of “top-down causation”. Externally imposed boundary conditions – such as the modulated carrying capacity K_n in our model – constrain the dynamics of all units from outside the system
>
> […]
>
> This distinction clarifies that the “top-down” signal in our model is not a claim of autonomous macroscopic agency, but rather a demonstration that externally structured environments can generate the informational preconditions from which internally coordinated systems may subsequently emerge.